

Large Language Model

LLM is an advanced machine learning model deep learning based that was provided a very large amounts of data focusing on words , paragraphs and linguistics to train from thousands of books, websites and articles ,relying on artificial neural networks (with thousands of adjustable parameters and weights) imitating the human brain,however, instead of looking for patterns and rules , it searches for similarities that can complete a sentence based on how familiar the order of words is to it.

On higher levels

As it is made clear that an LLM focus on words it doesn't it actually understand but goes through deeper layers as follows :

1. **Tokenization** : An LLM breaks words , symbols and characters into smaller components called tokens and each token is assigned to a **numerical id**.
2. **Embedding Layers** : Once it's converted into numbers it maps them to higher dimensional vectors and it searches for similar words classified previously while keeping order and structure of the sentence.
3. **Hidden Layers** : The vectors then travel through deeper layers that maps words using context like principle that evaluates percentage/score to find how relevant the meaning to the original text and reach higher levels of logic and relationships.
4. **Softmax Layer** : The final layer where one token is chosen based on probabilities.

5. **Generation Loop** : LLMs produce one token at time as the output, then appended back to the system to find the next token to output until it stops through a predefined length limit.

Popular LLMs & Applications

There are multiple LLMs that are available on the Internet but some of their most popular are like GPT-4 , Claude 3.5 Sonnet , Gemini 1.5 Pro.

Applications :

- **Coding & Software Development**: Writing codes and debugging errors to help developers complete more efficient tasks in less time.
- **Summarization**: Might summarise long forms and reports into bullet points .
- **Language Translation** :Translating texts and paragraphs from a language to another while maintaining meaning behind cultural expressions and traditions.